# Sandeep Kumar Amgothu

*Machine Learning Engineer · LLM Safety & Red Teaming · Data Engineering · MLOps*

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## Professional Summary

Machine Learning Engineer with 5+ years building production-grade AI systems specializing in **LLM security, adversarial red teaming, computer vision,** and **scalable ML infrastructure**. Architected automated LLM safety pipelines executing 3,000+ adversarial interactions, data/ML workloads processing **15M+ records/month**, and UAV detection models achieving **97.2% F1-score**. Published researcher (ICUAS 2025) bridging academic innovation with enterprise-grade AI safety engineering. **Work Authorization: STEM OPT Eligible (3 Years). No sponsorship required.**

## Core Technical Skills

**ML & AI:** PyTorch, TensorFlow, Scikit-learn, XGBoost, supervised/unsupervised learning, classification, anomaly detection
**LLM & GenAI:** GPT-4/4o, LLaMA-2/3, BERT, T5, instruction-tuning, LoRA/QLoRA, prompt engineering, few-shot learning, LLM-as-a-judge
**AI Safety & Security:** Automated red teaming, jailbreak generation, PYRIT Crescendo, taxonomy-driven evaluation, safety finetuning, Llama Guard, Detoxify
**MLOps & Infra:** MLflow, DVC, Weights & Biases, Docker, Kubernetes, CI/CD (GitHub Actions), A/B testing, reproducible pipelines
**Data Engineering:** Python, SQL, PySpark, Apache Spark, Airflow, ETL/ELT, AWS (S3, Glue, Lambda, Redshift, EMR), Snowflake
**Vector Search & RAG:** FAISS, Pinecone, Milvus, Chroma, LangChain, dense retrieval, semantic search, Elasticsearch
**Cloud & DevOps:** AWS, Azure, GCP (familiar), Terraform (familiar), Linux, REST APIs, serverless architectures
**Analytics & Viz:** Power BI, Tableau, Matplotlib, Seaborn, Plotly, Streamlit, KPI dashboards

## Professional Experience

**Texas A&M University–Corpus Christi (TAMUCC)** **Corpus Christi, TX**
*Graduate Research Assistant — AI/ML Research Engineer* Jan 2024 – May 2025
*Supervisor: Prof. Dulal Kar, Ph.D. (Dept. Chair) · CI 321 · (361) 825-5878 · dulal.kar@tamucc.edu*

- Spearheaded automation-first, taxonomy-driven LLM red-teaming framework executing **3,000+ adversarial interactions** across six model/attack configurations with database-backed logging for replayable safety analysis.
- Engineered closed-loop pipeline integrating GPT-4o-mini as red-team agent, quantized Llama-3.2-1B-Instruct as target, and ensemble safety graders (Llama Guard, DeBERTa, Detoxify, RoBERTa) into a composite **L0–L3 harm scoring engine**.
- Orchestrated three automated attack regimes (OpenAI baseline, multi-turn PYRIT Crescendo, jailbreak templates) exposing failure modes missed by naive evaluation and measuring **20–50 pp attack success reduction** via safety finetuning.
- Architected end-to-end MLOps pipeline for UAV audio detection: parallelized STFT/Mel spectrogram preprocessing for **50,000+ signals** with Spark/Python, slashing feature extraction from 15 hrs to under 5 hrs.
- Delivered **97.2% F1-score** and 90%+ accuracy for real-time UAV detection across 8 drone types with MLflow tracking and <100ms inference latency for edge deployment.
- Deployed production-oriented RAG systems (LangChain, FAISS, HuggingFace) achieving **sub-200ms query latency** for enterprise knowledge management.
- Published peer-reviewed research at **ICUAS 2025**; additional paper on automated LLM red teaming under review.

**Tata Consultancy Services (TCS)** **Hyderabad, India**
*Data Engineer* Nov 2021 – Jul 2023

- Architected cloud-native ETL pipelines on AWS (S3, Redshift, Glue, Lambda) processing **15M+ legal & financial records/month**, improving latency by **40%** through optimized partitioning and incremental loads.
- Delivered **99.9% data availability** for Fortune 500 client (Xerox), enabling real-time analytics for **500+ enterprise users** across legal and financial operations.
- Automated ingestion of **10K+ legal documents/month** via Python web scrapers (BeautifulSoup, Selenium) with NLP extraction, cutting manual processing by 35%.
- Elevated downstream ML model accuracy by **15%** via feature engineering pipelines and automated data quality

frameworks (schema validation, anomaly detection, profiling).

- Deployed executive-facing Power BI/Tableau dashboards with real-time KPI tracking, reducing report turnaround by **35%**.

**Accenture** **Hyderabad, India**

*Associate Data Engineer — AI/Analytics Trainee* Jun 2021 – Oct 2021

- Automated recurring analytics workflows (Python, SQL, Power BI), eliminating **25% of manual reporting** and cutting error rates by 30% via robust validation and standardized transformations.
- Engineered preprocessing and feature engineering scripts for datasets with **500K+ records**, improving ML/analytics performance by 15%.
- Delivered 5+ client analytics projects in Agile cross-functional teams, strengthening ETL best practices and stakeholder communication.

## Key Projects

**Automated Taxonomy-Driven Red Teaming for LLMs** | GitHub Jan 2025 – Present

- Designed enterprise-grade adversarial testing framework with persistent seed/prompt/response logging, taxonomy-aware grading, and six attack/model configurations for scalable LLM safety evaluation.
- Built weighted ensemble scorer (DeBERTa reward + Llama Guard + Detoxify + RoBERTa) producing composite harm metrics and L0–L3 risk scale, enabling per-category attack analysis and longitudinal defense tracking.
- Demonstrated **20–50 pp attack success reduction** via safety finetuning with especially strong gains on discrimination, information hazards, and malicious-use harms.

**UAV Audio Detection via Deep Learning** | GitHub Jun 2024 – Jan 2025

- Developed spectrogram-based CNN/RNN/CRNN/VGG19 models detecting unauthorized UAVs from acoustic signatures with **97.2% F1-score** across 8 drone types; published at ICUAS 2025.
- Scaled data augmentation pipelines (SpecAugment, pitch-shift, noise injection) via Spark/Python, boosting generalization by 12% and enabling <100ms real-time edge inference.

**Supply Chain Analytics Platform (Power BI & Python)** May 2025 – Aug 2025

- Delivered full-stack analytics solution integrating SQL Server, Python preprocessing, and automated Power BI dashboards monitoring inventory, delivery, and water-quality compliance in near real time for **200+ stakeholders**.
- Reduced manual reporting effort by ∼30% and enabled data-driven corrective actions on KPI deviations.

## Education

**Texas A&M University–Corpus Christi** — M.S., Computer Science Aug 2023 – May 2025

*GPA: 3.93/4.0* · Leadership Scholarship Recipient

Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, AI Safety, Distributed Systems, Cloud Computing

**Vasavi College of Engineering** — B.E., Information Technology Aug 2017 – Aug 2021

*GPA: 3.64/4.0*

## Certifications & Publications

- **Microsoft Certified: Power Platform Developer Associate** Feb 2026 – Feb 2027
- **Microsoft Certified: Power BI Data Analyst Associate** · ID: 6FAC44F797A833FA Jun 2025 – Jul 2027
- **Microsoft Certified: Azure Administrator Associate** · ID: A80D13DAD5259C2B Feb 2023 – Feb 2027
- *UAV Audio Detection and Identification Using STFT Spectrograms with Deep Learning*, **ICUAS 2025**
- *Automated Taxonomy-Driven Red Teaming for Large Language Models*, under review, 2025
- *UAV Detection Systems*, TAMUCC Student Innovation Symposium, 2025

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